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Scalable Extraction of Author-Stated Research Gaps: Towards Automated Idea Generation

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Abstract

Papers state their own research gaps: limitations, future work and open problems. These statements could seed automated idea generation, but they are rare, scattered and easily confused with criticism of prior work. Asking a large language model (LLM) to judge every sentence is costly, and giving it the whole paper buries a few relevant sentences. We instead narrow each paper with a funnel: structural rules discard unlikely text, a natural-language-inference classifier proposes candidates, and the LLM verifies them without rewriting. Every output is a sentence the authors wrote, and one instruction sets the verifier to favour precision or recall. We evaluate on human-labelled limitation (SAL) and future-work (FWS) benchmarks and on 230 papers labelled by two LLM annotators and an LLM judge. The instruction controlled the trade-off on every dataset, and verification more than doubled SAL precision at 85% recall. Because the LLM only checks a short candidate list, it reads about 7–8 times fewer input tokens than whole-paper prompting on full papers, while matching or improving F_1 on the human-labelled benchmarks at every model size. The gain is largest for small models: with a 4B LLM, F_1 rose from 0.30 to 0.48 on arXiv papers and from 0.66 to 0.71 on SAL.

Keywords: research gap extraction; limitations; future work; scientific text mining; large language models; natural language inference; zero-shot classification; LLM filtering

1. Introduction

Research papers often say where a study falls short and what should be done next. Authors acknowledge the limitations of their own work, which tell readers how far its design, execution and interpretation limit the findings [1,7]. They also propose future work, naming the extensions and open questions they consider worth pursuing [2,3], and they point to unsolved problems in their field when they motivate their own work. We call these statements *author-stated research gaps*. Collected across many papers, they are a natural input for automated idea generation, because each one is a domain expert's claim about what is still missing.

Gap statements are hard to find for four reasons. First, they are rare. In the test sets we use, they make up only 2.6% of the sentences of randomized controlled trial (RCT) reports, 5.7% of full arXiv and biomedical papers and 14.6% of conclusion sections. Second, they are scattered. A limitation may sit in a dedicated section, in the discussion or next to a methodological choice, and in computer-science papers many gaps appear in the methods

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and experiments sections. Third, many sentences look like gaps without being one, such as a criticism of prior work or a restated result. Finally, generating gaps is no shortcut for finding them. LLMs can be prompted to write limitations and critiques of a paper [4,5], but these express the model's view rather than the authors', whereas we want every output to be a sentence the authors actually wrote.

A large language model (LLM) can judge whether a sentence states a gap, so the open question is how much of the paper to give it. Judging every sentence in a separate request is expensive. Giving the model the whole paper is simpler, but the few relevant sentences then sit among thousands to tens of thousands of tokens. LLMs use information in the middle of long inputs less reliably [6], and their accuracy falls as inputs grow [34]. This matters most for the smaller models that make large collections affordable to process.

Our idea is to minimise the LLM's job. Instead of asking it to find the gaps in a paper, we ask it only to verify a short list of candidates that cheaper components have already selected. Earlier extraction work points in the same direction. Clinical limitation recognition selects candidate sentences with section titles and limitation keywords, and these rules alone were more accurate than supervised classifiers [1], while future-work recognition separates locating sentences from grouping or classifying them [2,3]. We build this division of labour into a three-stage funnel (Figure 1). Stage A removes text that is unlikely to contain gaps, Stage B selects candidate sentences with a zero-shot natural-language-inference (NLI) classifier, and Stage C asks an LLM to keep or drop each candidate. Because no stage rewrites text, every output is a sentence the authors wrote, and because Stage B needs no gap-labelled training data, the same funnel runs on biomedical and computer-science papers. As the LLM only verifies, a single instruction can also steer it. A mode paragraph in the system prompt sets Stage C to high precision, balanced or high recall. Since a false gap wastes effort in idea generation, precision is our primary goal and high precision is the main mode.

This design raises three questions, which structure the rest of the paper.

- **RQ1. How do rules, a classifier and an LLM each contribute when they extract research gaps together?** Because the LLM can only verify what the earlier stages pass on, we measure how much precision each stage adds and how much recall it costs. We also test whether a training-free NLI classifier is good enough for Stage B, compared with a classifier trained on human labels and one fine-tuned on our LLM labels, and whether the instruction alone can make the LLM stage stricter or more lenient.
- **RQ2. How does filtering a paper before an LLM reads it compare with giving the LLM the whole paper, in accuracy and cost?** This question tests our central idea. With the same LLM on both sides, we compare F_1, input tokens and time for 27B, 9B and 4B models, sending the funnel's candidates either 20 per call or all of a paper's candidates in one call.
- **RQ3. How far can labels made by LLMs stand in for human labels when evaluating gap extraction?** Human labels exist for only one gap type per dataset, so part of our evaluation relies on labels produced by LLMs. We answer this question on the datasets that have human labels, by scoring the same SAL output against both label sets and by testing a classifier fine-tuned on LLM labels against the human labels of SAL and FWS. The answer is therefore partial, and a human study of all three gap types is the next step.

To answer them, we evaluate the funnel on two benchmarks with human labels, self-acknowledged limitations in RCT reports (SAL) [7] and future-work sentences in natural language processing (NLP) papers (FWS) [3]. Because each covers only one gap type in one field, we also built a corpus of 230 arXiv and biomedical papers, in which two LLMs

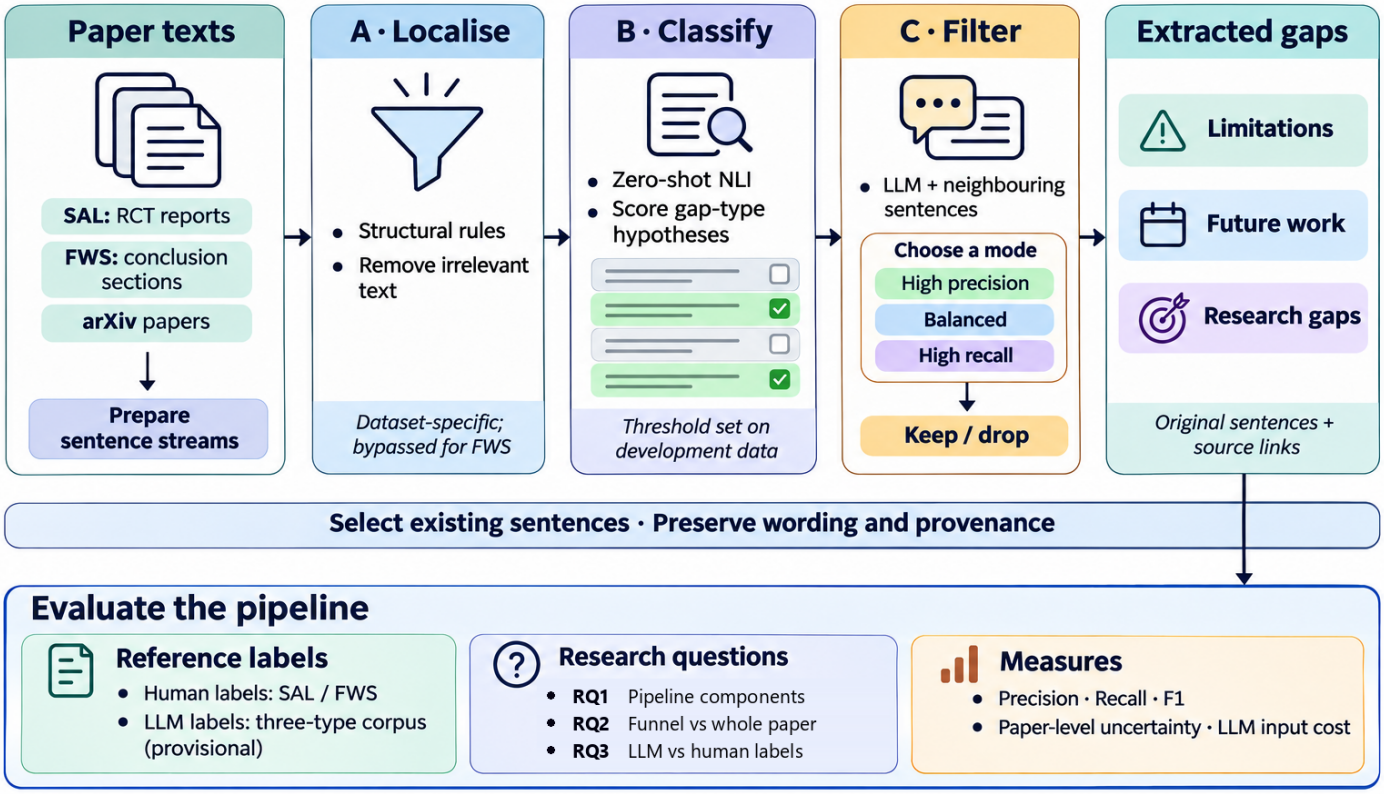


Figure 1. Overview of the funnel. Each paper is split into sentences. Stage A removes text that is unlikely to contain gaps (its rules depend on the dataset, and FWS skips it), Stage B selects candidate sentences with a zero-shot NLI classifier, and Stage C asks an LLM to keep or drop each candidate. The three filter modes are alternative instructions for Stage C, not consecutive steps. Every output is an original sentence with its source. Evaluation uses human labels on SAL (limitations) and FWS (future work), and provisional LLM labels for all three gap types.

labelled all three gap types independently and an LLM judge resolved their disagreements. We compare the funnel with whole-paper prompting using 27B, 9B and 4B models.

Narrowing the LLM's job pays off most where the LLM is weakest. With a 4B model, whole-paper prompting kept most sentences of several long arXiv papers. In contrast, the funnel raised F_1 on arXiv from 0.30 to 0.48 while the LLM read about seven times fewer input tokens. In this setting the funnel sends each paper's candidates in a single call and uses a Stage B classifier fine-tuned on our LLM labels (Tables 8 and 9). With a 27B model the two approaches are closer. The funnel scores higher on the human-labelled benchmarks with less input, while whole-paper prompting scores higher on arXiv papers, whose labels were themselves made from whole papers. Our contributions are as follows.

1. **A source-preserving funnel for gap extraction**, in which the LLM only verifies candidates and one instruction sets the trade-off between precision and recall. The modes kept their order on every dataset, and in high-precision mode verification raised precision on SAL from 29.1% to 66.2% at 84.9% recall (RQ1, Sections 4.1 and 4.3).
2. **A comparison with whole-paper prompting across three LLM sizes**, in both accuracy and cost. In its one-call-per-paper format, the funnel reads about 7–8 times fewer input tokens on full papers, and its advantage grows as the model gets smaller (RQ2, Sections 4.4 and 4.5).

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3. **An analysis of the stages and of the classifier.** Stage A helps where section headings are informative, and Stage B sets the recall ceiling because Stage C can only remove candidates. Stage C then supplies most of the precision gain on full papers. A zero-shot Stage B classifier comes within 0.021 F_1 of a supervised one on SAL, although it is less precise (RQ1, Sections 4.1 and 4.2).
4. **A three-type evaluation corpus and evidence on how far LLM labels can replace human ones.** We release the LLM-labelled corpus together with every LLM verdict behind our results. The same SAL output reaches 66.2% precision against human labels but 93.2% against LLM labels, so precision measured against LLM labels cannot be read as precision against human judgement. We also find that letter case reveals the label in the public FWS corpus, which is why we lowercase all FWS text (RQ3, Section 4.6).

The scope of this paper is extraction, so whether an extracted gap is still open, and whether it leads to a useful research idea, are questions we leave for later work. Section 2 reviews related work and Section 3 describes the data, the funnel and the comparisons. Section 4 reports the results, Section 5 discusses them together with the limitations of the study, and Section 6 concludes.

2. Related Work

Gap2Idea builds on research into scientific discourse classification, the recognition of author-stated limitations and future work, and LLM-based analysis of scholarly documents. Across these areas, similar terminology can refer to different tasks: locating an existing statement, inferring a missing piece of knowledge, or generating a new criticism or research direction. We organise the discussion around these distinctions and explain how each line of work informs our funnel. To broaden coverage, we used LR-AI [8] for a supplementary structured search of OpenAlex and arXiv records; candidate papers retained here were checked against their primary records. Table 1 compares the closest studies by target, method and input unit.

2.1. Scientific discourse and sentence classification

Scientific discourse annotation provides a foundation for identifying statements by their role within a paper. Argumentative zoning represents the rhetorical function of sentences and their relationship to the authors' argument [9]. Liakata et al. [10] distinguish this rhetorical perspective from Core Scientific Concepts (CoreSC), which represent components of an investigation, including its goals, methods, observations and conclusions. The Dr. Inventor corpus brings the discussion closer to our targets: its 40 computer-graphics articles have a Future Work category in the argumentative-structure layer and a Limitation category among the crosswise features [11]. These annotations establish a full-text precedent for representing both kinds of statement, although they do not constitute a single research-gap detection task.

Abstract sentence classification supplies further resources and modelling approaches. PubMed 200k RCT and CSAbstruct support the recognition of abstract roles in biomedical and computer-science publications, respectively [12,13]. MAZEA allows multiple rhetorical labels per sentence [14], while Emerald 110k extends coverage across disciplines and includes a combined Research limitations/implications heading [15]. Such resources provide useful discourse supervision, but their labels and document scope differ from ours: a conclusion sentence need not state a gap, and a combined limitations/implications label also covers other content.

More targeted rhetorical schemes come closer to our extraction targets. Sci-Sentence includes Research Gap and Limitation among seven categories intended to support literature-

review analysis [16]. Its benchmark samples sentences from introductions, related-work sections and limitations sections. Its Research Gap category overlaps with our open-problem target, whereas its Limitation definition concerns weaknesses of a previous study rather than exclusively the authors' own work. This difference in attribution matters when deciding which statements count as positives.

Context is another shared concern. Hierarchical sentence classifiers model surrounding discourse [17], and Brack et al. [18] investigate cross-domain sequential classification through multi-task learning. Lan et al. [19] extend sequential classification to LLMs and multi-label outputs, while Zhang and Zhang [20] use context-enhanced representations to identify problem and method sentences. These studies motivate considering neighbouring text when a sentence is ambiguous. In Gap2Idea, local context is supplied to the final LLM filter, which decides whether a previously selected candidate expresses the target statement.

2.2. Recognition of limitations and future work

Work on self-acknowledged limitations directly addresses one of our three targets. Kilicoglu et al. [1] formulate limitation recognition in clinical research as sentence-level binary classification and investigate both rules and learned classifiers. Their use of section headings and limitation zones provides a direct precedent for structural localisation. Lan et al. [7] subsequently annotate limitations in 200 RCT publications and separate detecting a limitation from assigning its type. Their fine-tuned PubMedBERT detector achieves a reported F_1 of 0.821 under their evaluation protocol. SAL supplies our human-labelled limitation benchmark, and the PubMedBERT model family supplies a supervised comparison. We distinguish that published score from our full-paper evaluation because the evaluated sentence pool and class balance affect precision and F_1.

Future-work recognition has a parallel history. Hu and Wan [21] use regular expressions to extract future-work statements, classify them into four categories and expose the results through a search prototype. Zhu et al. [2] use BERT-based recognition to identify future-research sentences in JASIST articles and cluster the extracted statements. The ACL FWS-RC corpus provides annotations and a taxonomy for future-work sentences in NLP papers [22]. Using this resource, Zhang et al. [3] recognise future-work sentences with feature-based classifiers, the strongest being Bernoulli naive Bayes (macro-F_1 of 0.9073), and classify their directions with neural models. Their recognition and direction-classification tasks are distinct, as are macro-F_1 and the positive-class F_1 used in our main evaluation. These published scores therefore provide context rather than a ranking against our paper-disjoint experiments.

Extracted statements also support analysis beyond detection. Suray et al. [23] examine how subsequent usable-security research engages with previously stated future work. More recently, Thaepprasit et al. [24] analyse self-reported limitations in ACL and EMNLP papers through iterative human–AI qualitative coding. At the downstream ideation stage, Si et al. [25] find in a controlled study with more than 100 NLP researchers that LLM-generated ideas were judged more novel but slightly less feasible than expert ideas. These studies show both the value of author-written statements and the need to evaluate extraction separately from the ideas produced from it. Gap2Idea addresses the preceding extraction step; it does not establish whether a stated gap remains unresolved or whether a proposed direction is useful.

2.3. LLM extraction, inferred gaps and generated critique

LLMs are increasingly used both to extract statements and to produce new assessments of research. BAGELS constructs limitation data from ACL, NeurIPS and PeerJ papers, sup-

plements author-written material with external reviews, and studies retrieval-augmented limitation generation [26]. FutureGen uses LLM-extracted author future-work statements, augmented with reviewer suggestions, as the reference for generating and evaluating new research directions [27]. Cheng et al. [28] directly investigate prompted LLM extraction of future-work sentences from the full text of NLP papers and connect the extracted material to keyword-based research-direction analysis. Together, these studies establish that LLM extraction of author-stated gaps already has close precedents.

Knowledge-gap identification also includes an inference task. GAPMAP explicitly distinguishes stated biomedical knowledge gaps from missing knowledge inferred from context, evaluating LLMs in paragraph-level and full-paper settings [29]. This distinction is central to our scope: we select an existing sentence that articulates a gap and preserve its wording. We do not ask the model to construct an implicit gap that has no corresponding author-written statement.

In limitation generation and scientific critique, the model writes the output rather than selecting existing sentences. LimGen studies suggestive limitation generation using 4068 ACL papers and their associated author-written limitations [30]. Despite its similar name, LimitGen is a separate benchmark for evaluating LLM-generated limitations and feedback on AI research papers [4]. Tree-of-Concerns focuses specifically on unstated limitations through structured multi-agent critique [5]. These tasks can produce useful criticisms beyond those acknowledged by the authors. Our LLM instead returns a keep-or-drop decision for each candidate sentence; the resulting extraction remains attributable to the original text.

2.4. Position of this work

Our Stage B builds on established methodology. Yin et al. [31] formulate zero-shot text classification as textual entailment, allowing a label to be expressed as a natural-language hypothesis. We apply this formulation to gap types using an existing NLI model. Here, zero-shot means that the classifier is not fitted on gap-detection training examples; the model has already been trained on NLI data, and our decision threshold is selected with labelled development data.

Ma et al. [32] provide the closest general information-extraction analogue: a small fine-tuned model filters examples and an LLM reranks the difficult subset. Their system routes by estimated difficulty, whereas our Stage B selects possible gap statements and Stage C makes a binary source-preserving decision. FrugalGPT [33] places such staged designs in the broader family of LLM cascades that trade cost against quality. Our cascade differs by combining rules, a non-generative classifier and one LLM rather than routing queries among several LLM services.

Our contribution is the evaluation of this particular combination with three precision–recall instructions. We examine how the stages interact, how the filter settings behave across datasets, and how the funnel compares with giving the same LLM the whole paper, in accuracy and cost, for LLMs of different sizes. Long inputs are a known difficulty for LLMs [6]; RULER likewise finds large performance drops with increasing context length across 17 models [34]. In long-context question answering, order-preserving retrieval can outperform full-context input while using fewer tokens [35]. Closer to our setting, Wu et al. [36] first extract key sentences from a long document and only then prompt an LLM to evaluate a summary, which lowers cost and raises agreement with human judgements. These tasks differ from sentence extraction, but they motivate testing whether candidate selection helps an LLM focus on sparse evidence. The evaluation covers both full papers and the section-level FWS resource, keeping their processing units and label sources explicit. Human-labelled limitation and future-work benchmarks support those

two targets; findings for the joint three-type task remain provisional because that corpus is LLM-labelled. The resulting evidence concerns source-preserving extraction; assessing the quality of research ideas derived from these statements remains a separate task.

3. Materials and Methods

This section defines the task and the data, describes the three stages of the funnel and how their settings were chosen, and then introduces the systems we compare against and the evaluation protocol.

3.1. Task

Given a paper as a sequence of sentences, the system selects the sentences in which the authors state a research gap. We distinguish three types:

- **Limitation**: a weakness, caveat, bias, constraint or assumption the authors acknowledge about their own work;
- **Future work**: an explicit direction the authors propose for future research;
- **Research gap**: an unsolved problem, missing capability or open challenge the authors point to in the field or in prior work.

The system never rewrites or summarises; each output is an unmodified source sentence. Evaluation is binary at the sentence level: a returned sentence is correct if the reference labels mark it as a gap of a type the benchmark covers, and the predicted type is not scored. SAL labels only limitations and FWS only future work, so on these benchmarks the funnel is asked for that single type.

3.2. Data

We use three datasets (Table 2): two public benchmarks with human labels, each covering one gap type, and a corpus of 230 papers that we labelled for all three types with LLMs. Within each dataset, splits are made by paper.

SAL.

SAL contains 200 RCT reports in which annotators marked the sentences that state a self-acknowledged limitation, split into 120 training, 40 development and 40 test papers [7, 37]. We call the test split SAL-40 and the development split SAL dev. We removed sentences of fewer than three words (none of the removed test sentences is a limitation) and otherwise evaluate on the full paper. To apply section rules in Stage A, we recovered each sentence's section by matching it to the PubMed Central BioC version of its article. The training split is used only to train the supervised classifiers in Section 3.5.

FWS.

The FWS recognition corpus contains sentences from the conclusion sections of ACL papers, each labelled as future work or not [3,22,38]. Each "paper" is a conclusion excerpt of about seven sentences, so FWS has no full-paper structure; we use it to test the funnel on a second domain and a second gap type. We removed every sentence that also occurs in the training data of an earlier classifier of ours, shuffled the papers once, and took the first 209 papers (1506 sentences) as the test sample and the next 418 as the development sample. All FWS text is lowercased before any processing, for every system, because in this corpus letter case reveals the label (Section 4.6.2).

Gold corpus.

To evaluate all three gap types together, we built the *gold corpus*: 230 papers and 68,719 sentences. It contains 190 arXiv computer-science and machine-learning papers and the

Table 1. Selected related work, ordered by year of the cited version. Target categories describe the relevant part of each study; other rhetorical labels are omitted where applicable. When multiple target categories apply, they are listed in the order Limitations, Future work and Research gaps. "Input unit" is the input each method processes.

Study	Domain	Target categories	Approach	Input unit	Relation to this study
Hu and Wan, 2015 [21]	Computer science	Future work	Rules; multi-class classification	Sentences, full articles	Early extraction, typing and search prototype
Dr. Inventor, 2016 [11]	Computer graphics	Limitations; Future work	Human multi-layer annotation	Sentences, full articles	Full-text precedent with both statement types
Kilicoglu et al., 2018 [1]	Clinical research	Limitations	Rules; supervised classifiers	Sentences	Precedent for section-based localisation
Zhu et al., 2019 [2]	Information science	Future work	BERT classification; clustering	Sentences, full articles	Links recognition to thematic analysis
ACL FWS-RC, 2020 [22]	NLP	Future work	Annotation; taxonomy	Sentences	Source of FWS labels
Zhang et al., 2023 [3]	NLP	Future work	Feature-based recognition; neural typing	Sentences	FWS recognition benchmark
Ma et al., 2023 [32]	General information extraction	IE labels	Small-model filter; LLM reranker	Instances	Closest filter-then-LLM analogue
LimGen, 2024 [30]	NLP	Limitations	LLM generation	Paper	Author limitation sections as references
Lan et al. (SAL), 2024 [7]	RCT reports	Limitations	PubMedBERT detection and typing	Sentences	Source of SAL; supervised comparison
Lan et al. (multi-label), 2024 [19]	Several sciences	Rhetorical roles	LLM prompting and fine-tuning	Abstract sequence	Context-aware LLM labelling
Zhang and Zhang, 2024 [20]	NLP	Problems; Methods	Context-enhanced transformer	Sentence with context	Local context for sparse targets
Brack et al., 2024 [18]	Several sciences	Rhetorical roles	Cross-domain multi-task learning	Sentence sequence	Transfer across domains
Suray et al., 2024 [23]	Usable security	Future work	Statement and citation analysis	Statement	Downstream use of extracted directions
BAGELS, 2025 [26]	NLP; machine learning; biology	Limitations	Extraction; retrieval-augmented generation	Paper; reviews	Extraction and generation benchmark
FutureGen, 2025 [27]	NLP; machine learning	Future work	LLM extraction; retrieval-augmented generation	Paper	Extraction as reference for generation
GAPMAP, 2025 [29]	Biomedicine	Knowledge gaps	LLM detection and inference	Paragraph to paper	Explicit versus inferred gaps
LimitGen, 2025 [4]	AI	Limitations	LLM critique; retrieval	Paper	Evaluates generated limitations
Sci-Sentence, 2026 [16]	Several disciplines	Limitations; Research gaps	Encoder and LLM classifiers	Sentence	Gap labels; different attribution
Tree-of-Concerns, 2026 [5]	Machine learning; NLP; vision	Limitations	Multi-agent critique	Paper	Unstated critique; different target
Cheng et al., 2026 [28]	NLP	Future work	Prompted LLM extraction; keyword analysis	Full text	Direct LLM extraction of future work
Thaepprasit et al., 2026 [24]	NLP	Limitations	Human–AI qualitative coding	Limitation sections	Downstream analysis of reported limitations
This study	RCT reports; NLP; computer science (arXiv)	Limitations; Future work; Research gaps	Rules, zero-shot NLI, LLM filter with three modes	Sentence with context; a paper's candidates	Stage contribution, precision control, comparison with whole-paper prompting

Table 2. Evaluation data. Positives are sentences labelled with a gap type the benchmark covers.

Dataset (split)	Labels	Gap types	Papers	Sentences	Positives	Use
SAL-40 (test)	Human	Limitation	40	7968	205	Main evaluation
SAL dev	Human	Limitation	40	7729	180	Threshold; held-out check
FWS (test)	Human	Future work	209	1506	220	Evaluation
FWS (dev)	Human	Future work	418	3005	418	Threshold
Gold corpus (test)	LLM ensemble	All three	34	11,741	664	Evaluation
Gold corpus (validation)	LLM ensemble	All three	34	11,201	699	Design choices; thresholds

SAL-40 papers with their original sentence boundaries. The arXiv papers are a convenience sample collected during earlier development of this project, not a random sample; their identifiers and extracted text are in the repository. They were converted from PDF with PyMuPDF and split into sentences after removing reference lists. We call the corpus "gold" because its labels are our reference for three-type evaluation, but no person has checked them yet, so all results on it are provisional.

The labels were produced in two steps:

1. *Independent labelling.* Two LLMs, Claude Opus 5.5 and Claude Sonnet 5.5, each labelled every sentence of every paper, in one call per paper, following a written guideline with the three definitions above. SAL labels were not shown to them. They agreed on the type of 2960 sentences.
2. *Adjudication.* The remaining 1737 candidates, marked by only one model or with different types, were judged again by Claude Opus 5.5 with three sentences of context on each side. It confirmed 1502 of them.

The corpus has 4462 gap sentences: 2234 research gaps, 1543 limitations and 685 future-work statements. The two labelling models agreed on whether a sentence is a gap with $\kappa = 0.778$ [39], which measures consistency between the models, not agreement with people. On the 40 SAL papers, the ensemble marked all but nine of SAL's 205 human-labelled limitations as a gap; this shows high recall against SAL, not precision.

Papers were split into 162 training, 34 validation and 34 test papers. The test split has 664 gaps (220 limitations, 97 future work, 347 research gaps); 28 of its papers are from arXiv (527 gaps) and 6 are SAL papers (137 gaps).

3.3. The funnel

The funnel applies three stages in sequence, each keeping a subset of the sentences it receives (Figure 1).

Stage A: localisation.

Stage A removes text that is unlikely to contain gaps before any model runs. Its rules depend on the dataset, because papers differ in how reliably their sections show where gaps are stated.

- **SAL:** only sentences in Discussion and Conclusion sections are kept.
- **Gold corpus:** no section rule is applied, because in arXiv papers many gaps are stated in the methods and experiments sections (Section 5.4). Stage A instead applies exclusion rules, which we call the *junk filter*. It drops titles, tables, figure captions and abbreviation lists; competing-interest, author-contribution, funding and supplementary sections where section types are known; bibliography-like strings; and malformed sentences (under 25 characters or six tokens, starting mid-word, containing dot leaders, or consisting mostly of non-letters).

- **FWS:** there is no Stage A. The corpus already consists of conclusion sections, and the junk filter, which rejects sentences that start in lower case, would remove almost all of the lowercased text.

Stage B: zero-shot NLI classifier.

Stage B selects candidates with a classifier that needs no gap-labelled training data. We use an NLI cross-encoder, ModernBERT-large fine-tuned on NLI data [40,41], as a zero-shot classifier [31]: the model reads a sentence (the premise) with a short statement of the label (the hypothesis) and estimates whether the sentence entails it. There is one hypothesis per gap type: "This is a limitation.", "This is future work." and "This is a research gap." For each hypothesis, the entailment and contradiction probabilities are renormalised to sum to one, and the entailment share is the score. On the gold corpus, a sentence becomes a candidate if its highest score over the three hypotheses reaches a threshold τ, and its predicted type is the hypothesis with that score. SAL uses only the limitation hypothesis and FWS only the future-work hypothesis. Inputs are truncated to 256 tokens.

Stage C: LLM filter.

Stage C keeps or drops each candidate with an LLM. Candidates are sent in reading order, 20 per request. For each candidate, the model receives its predicted type, the sentence (up to 400 characters) and a context window of the sentence and its immediate neighbours in the same paper (up to 600 characters). It returns a keep or drop verdict for every candidate, as JSON constrained by a schema. The system prompt (Appendix A) defines the three gap types, lists the kinds of sentences to drop (results, method descriptions, contribution claims, background, citations, captions and similar), and contains one paragraph that sets the mode. Only this paragraph changes between modes:

- **High recall**: when genuinely uncertain, keep the sentence;
- **Balanced**: favour neither keeping nor dropping;
- **High precision**: keep a sentence only if it explicitly states the gap itself; drop general background, motivation, observed outcomes and passing remarks; when uncertain, drop it.

For SAL and FWS, the definitions are narrowed to the benchmark's single type and the other two types are added to the sentences to drop. The mode paragraphs are identical on all datasets.

Our main model is Qwen3.8-27B (`Qwen/Qwen3.8-27B-FP8`, revision `017b9c7a`), which was not used to create any labels. It is self-hosted with vLLM 0.32.0 [42] on one H100 GPU, with temperature 0, seed 0, no reasoning mode and speculative decoding with two draft tokens.

3.4. How settings were chosen

To make clear which data could have influenced each setting, we list where it was chosen.

- **NLI hypotheses.** We compared hypothesis variants (long definitions, short type names, one combined hypothesis, adding neighbouring sentences to the premise, and subtracting competing "not a gap" hypotheses) on one half of the gold validation papers and confirmed the choice on the other half. Short type names worked best.
- **Thresholds.** τ is set on development data only, as the highest threshold at which recall after Stage B is at least 90%, counting positives removed by Stage A or Stage B as missed. This gives 0.7286 on SAL dev and 0.1922 on FWS dev. On the gold corpus, $\tau = 0.354$ was set on the same half of the validation papers that was used to choose the hypotheses; the rule applied to all 34 validation papers would give 0.232 and more

candidates. Stage B thus keeps most gaps and leaves the removal of false positives to Stage C.[1]

- **Stage C model, batch size and context.** These and the high-recall instruction were chosen during development on 196 candidates from 27 of the 34 gold test papers. We therefore also report gold results without these candidates.
- **Balanced and high-precision instructions.** These were written after reading false positives of a high-recall run on gold test papers, and checked on the gold validation split before those instructions were run on any test set. All three modes were run on every test set and all are reported, so no mode was selected by its test score. The frozen prompts and run outputs are in the repository history from 5 November 2026; earlier development is not time-stamped.
- **Held-out check.** SAL dev was used only to set τ; no prompt was written or chosen on it. Running the funnel on SAL dev therefore tests it on papers that influenced no prompt.

3.5. Comparison systems

Whole-paper prompting.

The main alternative to the funnel is to give the same LLM the whole paper. The baseline was implemented, and its plan, prompts and code frozen, before any test call. Each call contains one paper as numbered sentences in reading order (one per line, "[id] sentence"), with no Stage A or B, and the model returns the ids of all sentences it judges to be gaps. The system prompt is the funnel's Stage C prompt in the same mode, with only its opening line and the description of the input and output changed. Answers are limited to 4096 tokens; a call that exceeds the limit fails and its gaps count as missed. The arXiv sentence boundaries are those of the gold corpus, so both approaches are scored on exactly the same sentences.

On FWS, the frozen plan used a single older prompt from our code base that accepts limitations as well as future work. The FWS baseline therefore has one result for all three modes. This mismatch can understate its precision and prevents the FWS comparison from isolating the effect of the input strategy.

The SAL comparison is also asymmetric. The funnel uses section labels in Stage A, whereas the whole-paper baseline receives only numbered sentences. This favours the funnel on SAL.

The same frozen plan included two additional baselines: a chunked baseline using 100 consecutive sentences per call, and an every-sentence baseline using 20 consecutive sentences per call. The latter judges every sentence in the funnel's batch size but without Stages A and B. We report these as additional baselines for the 27B model.

Smaller LLMs.

To test how the comparison depends on model size, we repeat the funnel and the whole-paper baseline with Qwen3.5-9B (revision `c2022362`, bfloat16) and Qwen3.5-4B (`Qwen/Qwen3.5-4B`, revision `851bf6e8`, bfloat16). Both are served with vLLM using the same prompts and decoding settings as the 27B model but without speculative decoding. The 4B model sometimes returned 19 verdicts for a batch of 20 candidates; for such batches we asked again for the missing candidates alone, as the funnel's filter does, and no verdict remained missing. The 9B model completed every whole-paper call.

[1] The stored thresholds are rounded. At the rounded SAL value one development limitation falls below τ, so SAL dev recall before Stage C is 89.4% instead of 90.0%.

Supervised classifiers on SAL.

To test whether a training-free Stage B is enough, we compare it with classifiers trained on SAL's own labels. We trained eight classifiers on the SAL training split: three fine-tuned encoders, PubMedBERT [43] (`BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`), SciBERT [44] and DistilBERT [45], each trained for three epochs (learning rate 3×10^{-5}, batch 32, class-weighted loss) with a decision threshold chosen on SAL dev; and five frozen sentence encoders [46,47] with a logistic-regression classifier. PubMedBERT had the highest F_1 on SAL-40, and we use it as the supervised Stage B. Because this choice was made on the test split, it favours the supervised comparison.

Stage B fine-tuned on LLM labels.

We also fine-tuned the NLI model itself on the gold training split. Each training sentence is paired with each of the three hypotheses, labelled entailment when its gold type matches, neutral when it is a gap of another type and contradiction when it is not a gap. We trained on the 134 arXiv training papers only (33,404 sentences after Stage A, 2534 gaps), leaving out the 28 SAL-40 papers in the gold training split so that SAL-40 stays unseen. Settings were fixed before training: learning rate 10^{-5}, batch 32, two epochs, 6% warm-up, weight decay 0.01, seed 0, one V100 GPU. The epoch with the highest precision on the gold validation split at the 90%-recall rule was kept (the first). Thresholds follow the same rule on each dataset's development data.

Stage C call and answer format.

The funnel sends 20 candidates per call, each with its own context, and asks for a verdict on every one. We test two other call formats with the fine-tuned Stage B: one call per paper with all of the paper's candidates as numbered sentences (the call format of the whole-paper baseline, but with only the candidates), and the same with the paper's abstract placed before the candidates as context. Both return the ids of the sentences to keep. The arXiv abstracts were taken from the arXiv API and the SAL abstracts from the papers' abstract sections; FWS has no abstracts. Separately, we test the answer format alone: the funnel's 20-per-call requests, unchanged, but asking for the ids of the candidates to keep instead of a verdict on each.

Other Stage B options on the gold corpus.

We also compare the NLI classifier with bge-small, a small sentence-embedding model [47], with a logistic-regression classifier trained on the gold training split, and with the same NLI model using one long hypothesis that covers all gap types, used alone without Stage C (threshold 0.75, chosen on validation).

3.6. Evaluation

Metrics.

We report precision, recall and F_1 (their harmonic mean) for the positive class at the sentence level. Recall is computed over all positive sentences in the test papers, including those removed by Stage A, so the losses of every stage count. Sentences from the same paper are not independent, so we resample papers: confidence intervals (CIs) are 95% percentile intervals from 2000 bootstrap resamples of papers [48]. To compare two systems, we score both on the same resampled papers and report the CI of their difference, and call a difference significant when this interval excludes zero. The CIs are not corrected for the number of comparisons, so individual significant differences should be read as descriptive. Comparisons with whole-paper prompting on the gold corpus use its 28 arXiv test papers, on which the baseline was run.

Cost.

We measure cost as the number of LLM input and output tokens reported by the server, and as the summed time of the LLM calls, sent one at a time to one H100 GPU. The 27B whole-paper baseline ran on the second H100 GPU of the same server, in its own vLLM instance with the same model, revision and serving settings; its times are therefore only approximately comparable. The computation of Stages A and B, which use no LLM, is not included. The early 27B runs of the 20-per-call funnel did not log tokens; their input equals that of the logged 4B runs, because the prompts and tokenizer are the same.

Run-to-run stability.

To measure how much LLM verdicts vary between identical runs, we repeated the high-recall run on the gold test candidates.

Planned human study.

To measure precision against human judgement, we have designed a blind study. Two raters and an independent tie-breaker will label a frozen sample of 135 sentences, drawn from the high-precision output on arXiv and SAL-40 and from decoy sentences. Its results are not part of this version.

3.7. Software and use of generative AI

Experiments used Python with PyTorch, Hugging Face Transformers, sentence-transformers, scikit-learn and vLLM. The repository lock file pins the client environment. The LLM servers ran in a separate environment; the saved run records identify vLLM 0.32.0, but the client lock file does not reproduce the complete server environment. Generative AI is part of the method in two places: Claude Opus 5.5 and Claude Sonnet 5.5 produced the gold-corpus labels, and Qwen models are the Stage C filter and the whole-paper baseline. LR-AI [8] supported a supplementary search, deduplication and title-and-abstract screening for the related-work review; every newly cited source was checked against its primary record. The use of AI assistance in preparing the study and the manuscript is declared at the end of the paper.

4. Results

We report the results in the order of the research questions, followed by a study of the Stage C call format and two findings about the evaluation data. Unless stated otherwise, Stage C uses Qwen3.8-27B in high-precision mode, and results on the gold corpus use its provisional LLM labels.

4.1. Contribution of each stage (RQ1)

Each stage contributes differently; Stage C supplies most of the precision gain (Table 3 and Figure 2).

- **Stage A** removes many sentences only where sections are informative. On SAL-40, keeping only Discussion and Conclusion sections removes 76% of the sentences and loses 2 of the 205 limitations. On the gold corpus, the junk filter removes 18% of the sentences and 3% of the gaps.
- **Stage B** raises precision by a factor of 2.7 to 3.6 and keeps 88–91% of all positives, close to the 90% recall its threshold targets on development data.
- **Stage C** raises precision by a further 37 percentage points on SAL-40, 28 on FWS and 63 on the gold corpus, and lowers recall by 6, 12 and 34 points. On the gold corpus, most of the lost recall falls on research gaps and limitations (Section 4.3).

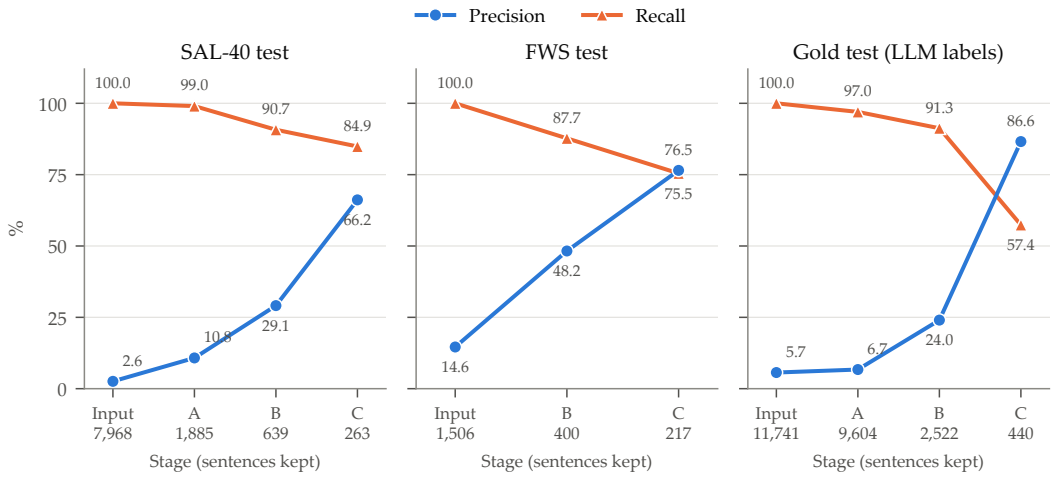


Figure 2. Precision and recall after each stage in high-precision mode. The number below each stage is the number of sentences it keeps. Gold results use LLM labels; FWS has no Stage A.

Table 3. Sentences kept after each stage. High-precision mode; precision and recall in %. Gold results use LLM labels.

Dataset	After stage	Sentences	Positives	Precision	Recall
SAL-40	Input	7968	205	2.6	100
	A. Discussion and Conclusion sections	1885	203	10.8	99.0
	B. NLI, $\tau = 0.7286$	639	186	29.1	90.7
	C. LLM filter	263	174	66.2	84.9
FWS test	Input (no Stage A)	1506	220	14.6	100
	B. NLI, $\tau = 0.1922$	400	193	48.2	87.7
	C. LLM filter	217	166	76.5	75.5
Gold test	Input	11,741	664	5.7	100
	A. Junk filter	9604	644	6.7	97.0
	B. NLI, $\tau = 0.354$	2522	606	24.0	91.3
	C. LLM filter	440	381	86.6	57.4

https://doi.org/10.3390/make1010000

4.2. Choice of Stage B (RQ1)

Supervised classifier.

We ran the funnel in all three modes with PubMedBERT, trained on SAL's own labels, as Stage B. Before Stage C, PubMedBERT selects 391 candidates with 46.0% precision and 87.8% recall (F_1 0.604); it was the best of the eight SAL-trained classifiers (SciBERT 0.553, DistilBERT 0.523, frozen encoders 0.371–0.457). After Stage C in high-precision mode it reaches 74.5% precision (95% CI 68.4–80.7), 78.5% recall and F_1 0.765.

With PubMedBERT, the funnel is 6.9 to 9.0 points more precise in every mode (Table 4). In high-precision mode the F_1 difference is small and not significant (-0.021, CI -0.053 to $+0.014$); in the other two modes PubMedBERT's advantage is larger (0.035 and 0.059). The training-free classifier is therefore competitive in F_1 in high-precision mode but not in precision. The NLI model's weights were not fitted on any SAL label; only its threshold was set on SAL dev.

Table 4. SAL-40, NLI versus PubMedBERT as Stage B. Left: the funnel with NLI minus the funnel with PubMedBERT in the same mode, paired over resampled papers; precision differences in points, 95% CI in brackets. Right: precision of the NLI funnel's output against the LLM labels instead of SAL's human labels (Section 4.6.1), in %.

	NLI minus PubMedBERT		**Precision against LLM labels**	
Mode	**Precision**	F_1	**Limitation**	**Any gap type**
High precision	−8.4 (−11.8, −4.5)	−0.021 (−0.053, +0.014)	93.2	95.1
Balanced	−6.9 (−11.5, −2.3)	−0.035 (−0.068, 0.000)	88.8	91.2
High recall	−9.0 (−12.7, −5.1)	−0.059 (−0.089, −0.027)	81.8	85.2

Fine-tuning on LLM labels.

Fine-tuning the NLI model on the gold training labels makes Stage B more precise against those labels but not against human labels. At the same 90%-recall rule, it cuts the gold test candidates from 2522 to 1998 and raises their precision from 24.0% to 30.4% at 91.4% recall. On SAL-40 and FWS it is slightly worse than the zero-shot model: 702 instead of 639 candidates at 26.6% instead of 29.1% precision on SAL-40, and 45.7% instead of 48.2% precision on FWS. After Stage C (Table 9), the fine-tuned funnel improves F_1 on the gold test split by 0.025 to 0.027 (significant in the high-recall and balanced modes) but not on SAL-40 or FWS (-0.019 to $+0.002$, none significant). One explanation is that the model learned how the LLM labellers mark gaps; differences in domain and input form between the training papers and the benchmarks may also contribute.

Other options on the gold corpus.

On the half of the gold validation papers not used to choose the hypotheses, the three short hypotheses reached 39.2% precision at about 79% recall, whereas the bge-small classifier trained on the gold training split reached 28.2% at the same recall. On the test split, the funnel in high-precision mode has higher F_1 than the NLI model used alone with one long hypothesis (0.690 against 0.614; difference 0.076, CI 0.020 to 0.132), and does not differ significantly from a funnel with the bge-small classifier as Stage B and the high-recall filter (F_1 0.665; difference $+0.025$, CI -0.037 to $+0.087$).

4.3. Precision control with three filter modes (RQ1)

Changing one paragraph of the instruction moves the filter along the precision–recall trade-off in the intended direction on every evaluation set (Table 5 and Figure 3): from high recall to balanced to high precision, precision rises and recall falls. In high-precision mode, precision is 66.2% on SAL-40, 76.5% on FWS and 86.6% on the gold test split (83.4% on its 28 arXiv papers and 97.1% on its six biomedical papers).

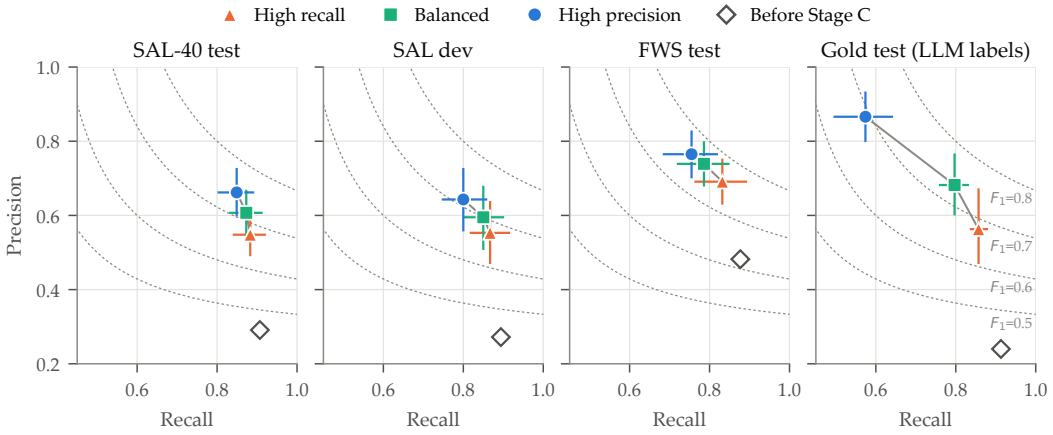


Figure 3. Precision against recall for the three filter modes and for the Stage B output before filtering, with 95% confidence intervals from resampling papers. Dotted curves connect points of equal F_1. Gold results use LLM labels.

Which mode has the highest F_1 depends on the dataset. On SAL-40, high precision has the highest F_1 (0.744, against 0.716 for balanced and 0.677 for high recall). On FWS, precision rises from 69.1% to 76.5% while recall falls from 83.2% to 75.5%, so F_1 stays within 0.007. On the gold test split, balanced has the highest F_1: 0.055 above high recall (paired 95% CI 0.025 to 0.080) and 0.045 above high precision (0.006 to 0.095).

By gap type.

High precision loses recall mainly on research gaps and limitations (Table 6). Future-work recall barely changes across modes (82.5–86.6%), whereas research-gap recall falls from 89.3% in high-recall mode to 51.3% in high-precision mode, and limitation recall from 79.5% to 55.9%. Research gaps are about half of the gold labels, so they weigh heavily on the overall scores: when sentences labelled as research gaps are excluded from scoring, high precision has the highest F_1 of the three modes (0.701, against 0.595 for balanced and 0.509 for high recall).

Checks.

The ordering of the modes also holds outside the test sets. On SAL dev, whose papers influenced no prompt, high-precision mode reaches 64.3% precision, 80.0% recall and F_1 0.713, close to SAL-40. On the gold validation split, where the modes were checked before the test runs, high-precision mode kept 60.3% of the labelled gaps among the Stage B candidates and rejected 97.7% of the other candidates, against 94.1% and 82.9% in high-recall mode. Removing the 196 gold test candidates used during development lowers F_1 slightly in every mode, from 0.690 to 0.675 in high-precision mode, 0.735 to 0.723 in balanced mode and 0.679 to 0.664 in high-recall mode. Removing the 27 papers they came from would leave only 7 test papers, too few to evaluate. Repeating the high-recall run with the same inputs and seed changed 32 of the 2522 verdicts (1.3%); F_1 was 0.678 in one run and 0.679 in the other.

Table 5. Results of the three filter modes. Precision and recall in %, with 95% confidence intervals from resampling papers. Gold results use LLM labels. Kept: number of sentences returned.

Evaluation set	Mode	Kept	Precision	Recall	F_1
SAL-40 test (human labels)	Before Stage C	639	29.1 (26.2–32.1)	90.7 (86.7–94.5)	0.441 (0.404–0.475)
	High precision	263	66.2 (59.7–72.8)	84.9 (80.1–89.3)	0.744 (0.695–0.789)
	Balanced	295	60.7 (54.6–66.9)	87.3 (83.0–91.4)	0.716 (0.670–0.760)
	High recall	330	54.8 (49.0–61.3)	88.3 (83.9–92.2)	0.677 (0.627–0.725)
SAL dev (human labels)	Before Stage C	591	27.2 (23.3–31.5)	89.4 (85.1–93.5)	0.418 (0.369–0.468)
	High precision	224	64.3 (55.7–72.8)	80.0 (74.6–86.0)	0.713 (0.647–0.774)
	Balanced	257	59.5 (50.7–68.0)	85.0 (79.9–90.2)	0.700 (0.628–0.764)
	High recall	282	55.3 (46.9–63.9)	86.7 (81.6–91.7)	0.675 (0.602–0.742)
FWS test (human labels)	Before Stage C	400	48.2 (42.2–54.5)	87.7 (81.1–93.4)	0.623 (0.565–0.677)
	High precision	217	76.5 (70.0–82.9)	75.5 (68.3–82.1)	0.760 (0.705–0.807)
	Balanced	234	73.9 (67.8–80.0)	78.6 (71.8–85.0)	0.762 (0.709–0.808)
	High recall	265	69.1 (62.9–75.3)	83.2 (76.2–89.4)	0.755 (0.702–0.804)
Gold test, all (LLM labels)	High precision	440	86.6 (79.8–93.4)	57.4 (49.4–64.3)	0.690 (0.619–0.748)
	Balanced	776	68.2 (60.0–76.7)	79.7 (75.8–83.3)	0.735 (0.678–0.789)
	High recall	1,011	56.3 (46.9–67.3)	85.7 (83.5–88.1)	0.679 (0.604–0.756)
Gold test, arXiv (LLM labels)	High precision	337	83.4 (76.0–91.3)	53.3 (44.4–61.1)	0.650 (0.573–0.713)
	Balanced	643	64.2 (56.0–73.6)	78.4 (73.8–82.6)	0.706 (0.647–0.765)
	High recall	862	51.7 (42.8–62.7)	84.6 (82.1–87.2)	0.642 (0.567–0.719)
Gold test, biomedical (LLM labels)	High precision	103	97.1 (94.5–99.0)	73.0 (63.7–81.0)	0.833 (0.761–0.885)
	Balanced	133	87.2 (82.4–92.2)	84.7 (80.5–87.7)	0.859 (0.833–0.889)
	High recall	149	82.6 (80.2–86.1)	89.8 (85.7–94.4)	0.860 (0.839–0.894)

Table 6. Gold test split by gap type (LLM labels). Recall for each type, and scores when research-gap sentences are excluded from scoring; in % except F_1.

Mode	Recall by type			Without research gaps		
	Limitation (220)	Future work (97)	Research gap (347)	Precision	Recall	F_1
High precision	55.9	82.5	51.3	77.5	64.0	0.701
Balanced	71.4	84.5	83.6	49.2	75.4	0.595
High recall	79.5	86.6	89.3	36.9	81.7	0.509

4.4. Funnel versus whole-paper prompting (RQ2)

The natural alternative to the funnel is to give the LLM the whole paper. Table 7 and Figure 4 compare the two with the same model and mode. These comparisons were designed after the main results were known. They are also asymmetric: the SAL funnel uses a section gate that the baseline does not receive, the FWS baseline uses a broader prompt than the funnel, and the arXiv labels were produced from whole-paper inputs. We therefore treat the differences as descriptive of these particular setups rather than as isolated effects of the input strategy.

Table 7. Funnel versus whole-paper prompting. F_1 of each with the same LLM and mode, and their paired difference (funnel minus whole paper) with its 95% CI; bold: the CI excludes zero. The intervals are not corrected for multiple comparisons. On FWS the whole-paper baseline has one broader prompt for all modes (Section 3.5). arXiv results use LLM labels.

Set	Mode	Qwen3.8-27B			Qwen3.5-9B			Qwen3.5-4B		
		Whole	Funnel	Funnel − whole	Whole	Funnel	Funnel − whole	Whole	Funnel	Funnel − whole
arXiv (28 papers, LLM labels)	High precision	0.679	0.650	−0.029 [−0.10, +0.04]	0.313	0.599	+0.286 [−0.03, +0.42]	0.296	0.544	**+0.248** [+0.07, +0.35]
	Balanced	0.749	0.706	−0.043 [−0.09, +0.02]	0.236	0.564	**+0.328** [+0.04, +0.41]	0.314	0.524	**+0.210** [+0.04, +0.32]
	High recall	0.741	0.642	**−0.099** [−0.16, −0.03]	0.403	0.556	+0.153 [−0.06, +0.25]	0.275	0.480	**+0.205** [+0.01, +0.27]
SAL-40 (human labels)	High precision	0.702	0.744	+0.042 [−0.01, +0.10]	0.684	0.690	+0.007 [−0.07, +0.09]	0.661	0.650	−0.012 [−0.08, +0.06]
	Balanced	0.673	0.716	+0.043 [−0.01, +0.09]	0.680	0.684	+0.004 [−0.07, +0.09]	0.618	0.611	−0.007 [−0.09, +0.07]
	High recall	0.663	0.677	+0.014 [−0.04, +0.06]	0.665	0.607	−0.059 [−0.15, +0.03]	0.599	0.576	−0.023 [−0.10, +0.04]
FWS (human labels)	High precision	0.694	0.760	**+0.065** [+0.02, +0.11]	0.654	0.731	**+0.077** [+0.03, +0.12]	0.719	0.727	+0.009 [−0.04, +0.07]
	Balanced	″	0.762	**+0.068** [+0.02, +0.12]	″	0.725	**+0.071** [+0.03, +0.11]	″	0.711	−0.008 [−0.06, +0.05]
	High recall	″	0.755	**+0.060** [+0.02, +0.11]	″	0.728	**+0.074** [+0.04, +0.11]	″	0.731	+0.013 [−0.04, +0.06]

With the 27B LLM.

With Qwen3.8-27B, the funnel has higher observed F_1 than whole-paper prompting on both human-labelled benchmarks. On SAL-40 the difference is 0.014 to 0.043 across modes, and every confidence interval includes zero. In high-precision mode the funnel reaches 66.2% precision and 84.9% recall; whole-paper prompting reaches 64.1% and 77.6%. On FWS the difference is 0.060 to 0.068 and its confidence interval excludes zero in every mode. The broader FWS baseline prompt, however, means these intervals do not isolate the effect

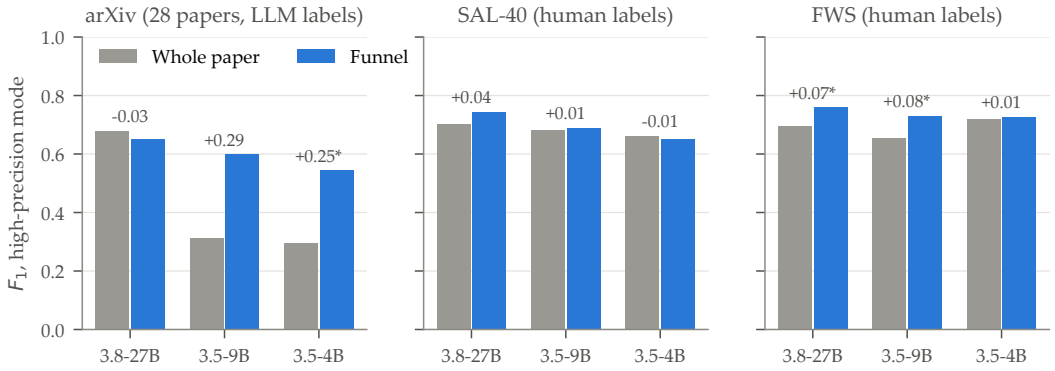


Figure 4. F_1 in high-precision mode for whole-paper prompting (grey) and the funnel (blue) with each LLM. Numbers give the difference (funnel minus whole paper); an asterisk marks a paired 95% CI that excludes zero. arXiv results use LLM labels.

of the funnel. On the arXiv papers the order reverses: whole-paper prompting is better in every mode, significantly in high-recall mode (-0.099). The other ways of splitting the paper show the same ranking. Chunks of 100 sentences reach F_1 0.70–0.73 on arXiv and 0.60–0.68 on SAL-40, and batches of 20 consecutive sentences reach 0.70–0.72 and 0.55–0.64. Both are ahead of the funnel on arXiv and behind it on SAL-40, while using more input tokens than the whole paper.

With the 9B LLM.

Qwen3.5-9B completes every whole-paper call but over-selects on several long arXiv papers, keeping 806 of 824 sentences in one mode and 499 of 692 in another. Across modes, its whole-paper F_1 is 0.236–0.403, against 0.556–0.599 for the main funnel. The funnel difference is 0.153–0.328; its confidence interval excludes zero only in balanced mode because the failures are concentrated in a few papers. On SAL-40 the observed difference ranges from -0.059 to 0.007 and no interval excludes zero. On FWS it is 0.071–0.077 and every interval excludes zero, subject to the broader baseline-prompt caveat.

With the 4B LLM.

The tested 4B model benefits from the funnel on long papers. With Qwen3.5-4B, whole-paper prompting performs poorly on arXiv papers: in high-precision mode it reaches 20.7% precision and 52.2% recall (F_1 0.296). On long papers the model returns most of the paper, for example 490 of 692 sentences. On the longest paper (1526 sentences, 83 gaps) its answer exceeded the output limit in every mode, but this is not the main cause: without that paper, F_1 is still 0.29–0.33. The funnel, which never sends the model more than 20 candidates at a time, reaches F_1 0.544 with the same model, an advantage of 0.21 to 0.25 whose confidence interval excludes zero in every mode. On SAL-40 and FWS, whose inputs are shorter, the two approaches do not differ significantly.

Paper length.

Splitting the arXiv papers at their median length (239 sentences) shows the same pattern within one dataset. On the 14 shorter papers, whole-paper prompting is ahead in high-precision mode with both models (F_1 0.707 against 0.624 for the 27B model, 0.658 against 0.595 for the 4B model). On the 14 longer papers, the observed scores are similar with the 27B model (0.668 for the funnel and 0.659 for the whole paper) and differ substantially with the 4B model (0.516 and 0.203). Within these models and papers, the funnel's benefit increases with input length and is larger for the 4B model.

Cost.

The funnel uses about a third of the input tokens of whole-paper prompting on SAL-40 and FWS. On arXiv, whole-paper prompting uses 1.4 times the funnel input because each of the 2137 candidates carries its own context (Table 8). The funnel's output is larger because it returns a verdict for every candidate rather than a list of kept sentences; with the 9B and 4B models this makes the 20-per-call funnel slower than whole-paper prompting on arXiv (287 against 92 seconds and 198 against 160 seconds). The chunked and every-sentence baselines are more expensive than the whole paper: with the 27B model they used 0.58–0.76 million input tokens on arXiv and 0.36–0.48 million on SAL-40. Section 4.5 shows a call format that removes most of the funnel's cost.

Table 8. Measured LLM cost in high-precision mode: input tokens (the same prompts for every model), output tokens and summed call time in seconds, one call at a time on one H100 GPU. arXiv: the 28 arXiv test papers; for the 20-per-call funnel, whose batches mix arXiv and SAL papers, each batch is counted by its share of arXiv candidates. The one-call-per-paper funnel uses the fine-tuned Stage B (Section 4.5). The early 27B runs of the 20-per-call funnel did not log output or time (n/a).

			Qwen3.8-27B		**Qwen3.5-9B**		**Qwen3.5-4B**	
Set	**Setup**	**Input**	Output	Seconds	Output	Seconds	Output	Seconds
arXiv	Whole paper	554k	1.6k	104	5.3k	92	6.0k	160
	Funnel, 20 per call	403k	n/a	n/a	23k	287	23k	198
	Funnel, one call per paper	75k	1.4k	31	2.1k	32	2.1k	24
SAL-40	Whole paper	348k	1.5k	65	1.1k	37	1.5k	30
	Funnel, 20 per call	106k	n/a	n/a	7.4k	89	6.4k	54
	Funnel, one call per paper	42k	1.1k	27	0.9k	21	0.9k	20
FWS	Whole paper	183k	2.7k	89	1.9k	80	2.1k	72
	Funnel, 20 per call	62k	n/a	n/a	4.0k	49	4.0k	34
	Funnel, one call per paper	67k	1.3k	59	1.2k	51	1.1k	47

4.5. Stage C call and answer format (RQ2)

Table 9. Funnel variants. F_1 of whole-paper prompting and of three funnels, by LLM; each cell gives high precision / balanced / high recall. On FWS the whole-paper baseline has one prompt for all modes. arXiv results use LLM labels.

Set	**Setup**	**Qwen3.8-27B**	**Qwen3.5-9B**	**Qwen3.5-4B**
arXiv	Whole paper	0.679 / 0.749 / 0.741	0.313 / 0.236 / 0.403	0.296 / 0.314 / 0.275
	Funnel (zero-shot NLI, 20 per call)	0.650 / 0.706 / 0.642	0.599 / 0.564 / 0.556	0.544 / 0.524 / 0.480
	Fine-tuned NLI, 20 per call	0.687 / 0.734 / 0.674	0.615 / 0.603 / 0.591	0.591 / 0.595 / 0.514
	Fine-tuned NLI, one call per paper	0.642 / 0.687 / 0.691	0.591 / 0.630 / 0.607	0.484 / 0.460 / 0.530
SAL-40	Whole paper	0.702 / 0.673 / 0.663	0.684 / 0.680 / 0.665	0.661 / 0.618 / 0.599
	Funnel (zero-shot NLI, 20 per call)	0.744 / 0.716 / 0.677	0.690 / 0.684 / 0.607	0.650 / 0.611 / 0.576
	Fine-tuned NLI, 20 per call	0.725 / 0.713 / 0.674	0.669 / 0.637 / 0.601	0.660 / 0.611 / 0.565
	Fine-tuned NLI, one call per paper	0.720 / 0.705 / 0.673	0.712 / 0.705 / 0.700	0.711 / 0.708 / 0.697
FWS	Whole paper	0.694 / 0.694 / 0.694	0.654 / 0.654 / 0.654	0.719 / 0.719 / 0.719
	Funnel (zero-shot NLI, 20 per call)	0.760 / 0.762 / 0.755	0.731 / 0.725 / 0.728	0.727 / 0.711 / 0.731
	Fine-tuned NLI, 20 per call	0.759 / 0.764 / 0.736	0.706 / 0.728 / 0.724	0.713 / 0.693 / 0.703
	Fine-tuned NLI, one call per paper	0.749 / 0.736 / 0.723	0.743 / 0.740 / 0.741	0.725 / 0.721 / 0.741

One call per paper.

Sending all of a paper's candidates in one call, numbered like the whole-paper baseline, reduces the input to one-seventh of whole-paper prompting on arXiv and about one-eighth

on SAL-40, with no more output. Whole-paper prompting also takes 1.5 to 7 times as much LLM time (Table 8). Accuracy depends on the model (Table 9). With the 27B model the one-call format differs from the 20-per-call format by -0.001 to -0.008 on SAL-40, and is 0.01 to 0.05 lower on FWS and in two of the three arXiv modes. With the 9B model it reaches F_1 0.591–0.630 on arXiv, 0.700–0.712 on SAL-40 and 0.740–0.743 on FWS; relative to whole-paper prompting, the interval excludes zero in arXiv balanced mode and every FWS mode. With the 4B model it is the best or tied-best setup on the human-labelled benchmarks: on SAL-40 it is ahead of whole-paper prompting by 0.050 to 0.098 (significant in the balanced and high-recall modes), with higher precision (67.1% against 56.8% in high-precision mode). Placing the paper's abstract before the candidates did not help; it lowered F_1 by up to 0.02 in every mode with the 27B model.

Answer format.

Asking for the list of candidates to keep instead of a verdict on each, with everything else unchanged, cuts output tokens four- to sevenfold but changes 7.6–9.3% of the verdicts, whereas repeating identical calls changes 0.4%. The model keeps more candidates in this format: precision in high-precision mode falls from 86.6% to 79.2% on the gold test split, at a similar F_1 (differences of -0.023 to $+0.006$, none significant).

4.6. LLM labels versus human labels (RQ3)

We answer RQ3 on the datasets that have human labels. On the 40 SAL test papers, the LLM labels include all but nine of SAL's 205 human-labelled limitations (Section 3), so they rarely miss what the annotators marked. They also mark many sentences that SAL leaves unlabelled, and this difference dominates the measured precision, as the first check below shows. A classifier fine-tuned on the LLM labels follows the same pattern: it becomes more precise against those labels but slightly less precise against the human labels of SAL and FWS (Section 4.2). The second check concerns the human labels themselves.

4.6.1. Precision depends on the label source

The precision of the same output changes greatly with the reference labels. Of the 263 sentences returned in high-precision mode on SAL-40, SAL's annotators marked 174 as limitations, whereas the LLM ensemble marked 245 as limitations and 250 as a gap of some type. The two label sets disagree mainly on 77 sentences that are gaps according to the LLM labels but not according to SAL; only 12 returned sentences are negative under both. Precision is therefore 66.2% against SAL and 93.2–95.1% against the LLM labels (right side of Table 4). Which value is closer to the truth depends on who is right about these 77 sentences, which the planned human study is designed to settle.

4.6.2. Letter case reveals the label in FWS

In the FWS recognition corpus as we obtained it (64,896 sentences from 9013 papers), all 9009 positive sentences are entirely lowercase, and almost no negative sentence is. The rule "a sentence is positive if it is all lowercase" therefore reaches 99.8% precision and 100% recall. Our copy has the same size in bytes as the published file [38], so the artefact appears to be present in the released corpus, although we did not compare the two files directly. Any case-sensitive model, and any LLM that sees the raw text, can exploit this signal. All FWS results in this paper therefore use lowercased text, and we do not use an earlier FWS result of ours that was obtained on raw text.

5. Discussion

We first interpret the answers to the research questions. We then discuss when whole-paper prompting is enough, how reliable the extracted gaps are, why section rules do not

transfer to computer-science papers, and what the results imply for evaluation data. The section ends with the limitations of the study.

5.1. Answers to the research questions

RQ1: each stage has its own role.

Stage A is cheap and useful only where section headings are informative. Stage B sets the ceiling on recall, because Stage C only removes candidates and cannot recover a gap that Stage B missed. Stage C provides most of the gain in precision. Every reported gap remains a sentence the authors wrote.

RQ1: a training-free Stage B is viable but less precise.

A zero-shot NLI classifier needs no gap-labelled data and works on biomedical and computer-science text. On SAL, a supervised classifier was 6.9 to 9.0 points more precise in every mode, while the high-precision F_1 difference was small and not significant. This supervised comparison is available only for SAL, and its classifier was selected on the SAL test split. Fine-tuning on LLM labels did not substitute for human labels.

RQ1: one instruction sets the observed trade-off.

The three modes kept their order on every evaluation set, on held-out SAL dev papers and on the gold validation split before any test run. Within these datasets and prompts, the instruction therefore provides a practical control: high precision for idea generation, where every false gap costs a downstream model or a person time, and high recall for a reviewer who wants to see every candidate. High precision costs recall mostly on research gaps. A possible reason, which we have not tested, is that the high-precision instruction asks the model to drop background and motivation, and authors often state open problems in exactly that role.

RQ2: the funnel helps the tested 9B and 4B models on long papers.

With the 27B model, the funnel's observed F_1 was higher on the two human-labelled benchmarks and lower on arXiv papers against LLM labels. The SAL differences were not significant, its section gate favours the funnel, and the FWS prompt mismatch prevents a clean attribution to the funnel. The funnel used about one-third of the input tokens on SAL-40 and FWS. With both Qwen3.5-9B and Qwen3.5-4B, whole-paper prompting performed poorly on long arXiv papers and the funnel did not. The shared Qwen3.5 generation shows that this pattern is not unique to the 4B checkpoint, but the 27B model belongs to the newer Qwen3.8 generation, so the three-way evidence still does not isolate parameter count from model design. The call format matters for cost as well. Sending each paper's candidates in a single call cuts the input on full papers to about one-seventh of whole-paper prompting on arXiv and one-eighth on SAL-40.

RQ3: LLM labels are not yet a substitute for human labels.

On SAL, the LLM labels include nearly every human-labelled limitation, but they also mark many sentences the annotators left unlabelled. The same output therefore scores 66.2% precision against human labels and 93.2% against LLM labels, and a classifier fine-tuned on LLM labels improves against those labels but not against human ones. Within these data, precision measured against LLM labels should not be read as precision against human judgement, which is why we call the three-type results provisional. Whether the same holds for future work and research gaps is what the planned human study will test.

5.2. When is whole-paper prompting enough?

Three factors decide the comparison in our data.

- **Input length.** SAL papers have about 200 sentences and FWS excerpts about seven; arXiv papers have a median of 239 and up to 1526, with whole-paper inputs of up to 72,000 tokens. The 9B and 4B models handled the short inputs but over-selected on several long arXiv papers. In the median-split analysis run for the 27B and 4B models, the funnel helped the 4B model on the longer half and not on the shorter half.
- **Model size and generation.** The 27B model handled every input, whereas the same-generation 9B and 4B models showed similar long-input failures. This makes it unlikely that the long-input failures are specific to the 4B checkpoint. The 27B model belongs to Qwen3.8 rather than Qwen3.5, however, so size is not the only difference between it and the two smaller models.
- **Reference labels.** The arXiv labels were produced by LLMs that read each whole paper in one call, the same input format as the whole-paper baseline, and may therefore favour it. On the two benchmarks labelled by people, the funnel was ahead with the 27B model.

The comparison is not perfectly matched in two further ways. On SAL, Stage A uses section labels that the whole-paper baseline does not see. On FWS, the baseline used an older prompt that also accepts limitations. Both favour the funnel; the arXiv label provenance favours the baseline.

The call format also matters. The funnel's main cost on arXiv is that each candidate carries its own context and receives its own verdict. Sending each paper's candidates in a single call removes most of this cost. The one-call-per-paper format is strong with the 9B model on all three datasets and is the most accurate setup on SAL-40 with the 4B model. With the 27B model, however, it loses up to 0.05 F_1 on arXiv, and asking for a list of kept sentences instead of a verdict on each candidate makes the model less selective. Cheaper formats are a trade-off rather than a free improvement.

5.3. How reliable are the extracted gaps?

How good the extracted gaps are depends on which labels are trusted. On SAL-40, high-precision mode reaches 66.2% precision against SAL's labels and 93.2% against the LLM labels. Either label set may be wrong on the 77 sentences where they disagree. SAL's annotators may have left some limitation statements unlabelled. The LLM labels are inclusive by construction, because adjudication added sentences that only one model had marked. The two values describe the same output under two references; neither is a bound on its true precision.

Label errors may also affect the gold corpus. In an earlier high-recall run during development, not part of the results above, five of the 34 test papers accounted for 59% of the false positives. Each had only about 18–24 gold gaps despite having 220 to 824 sentences, which may indicate that long papers are under-labelled. If so, recall on long papers is understated for every system, and precision against these labels is understated most for systems that keep many sentences.

The planned human study (Section 3) will measure precision against human judgement on a frozen sample from arXiv and SAL-40.

5.4. Section rules do not transfer to computer-science papers

In RCT reports, limitations cluster in the Discussion and Conclusion sections, so a section rule works. In arXiv papers it did not, in our development work. An earlier localiser kept only the sections where gaps are usually stated, such as the discussion and the conclusion. Measured against the limitation and future-work labels of the arXiv papers, with a fuzzy sentence match, it kept only 55.6% of these gaps when sections were inferred from the extracted text and 55.2% when sections were taken from the LaTeX sources of 175

papers. Better section information did not help: with LaTeX sections, 82% of the missed gaps were in methods, experiments or results sections. This development analysis used a different target and matching rule from our main evaluation and was not repeated for this version. For arXiv papers we therefore use only the junk filter as Stage A, which keeps 97% of the gold test gaps, and leave the narrowing to Stage B.

5.5. Evaluation data needs checking

The FWS case artefact shows that a published benchmark can contain a trivial signal that a modern model will find. FWS results from case-sensitive models, or from LLMs given the raw text, including an earlier result of ours, should be treated with caution unless the text is normalised. More generally, the precision we measure on SAL-40 changed more with the label source than with the choice of Stage B. Stating which labels a number rests on is as important as reporting the number.

5.6. Limitations

- **LLM labels.** All results on the gold corpus rest on labels produced by Claude models, which have not yet been checked by people. The agreement between the two labelling models ($\kappa = 0.778$) measures their consistency, not their accuracy. Claude models also assisted in writing the prompts and this manuscript, so the labels and the prompts may share biases.
- **Development on test papers.** The Stage C model, batch size, use of context and high-recall instruction were chosen on 196 candidates from 27 of the 34 gold test papers, and the balanced and high-precision instructions were written after reading false positives on gold test papers. We report results without these candidates, ran and report all modes on every dataset, and checked the modes on the validation split first; still, the gold test split is not untouched. SAL dev provides the cleanest held-out check, and its results agree with those on SAL-40.
- **Comparisons designed later.** The comparisons with whole-paper prompting, a smaller model and other formats were designed after the main results were known. The whole-paper baseline was frozen before its test runs, and all runs of these comparisons are reported.
- **Statistics.** Apart from one repeated high-recall stability check, each configuration was run once, and the many paired comparisons are not corrected for multiplicity. Small differences, especially those whose confidence intervals include zero, should not be treated as established effects.
- **Scope.** SAL covers RCT reports only, FWS covers NLP conclusion sections only, and each labels a single gap type. The six biomedical papers in the gold test split are also SAL papers. The arXiv papers are a convenience sample of 28 test papers. Stage C was tested with one model family.
- **Research gaps.** Half of the gold labels are research gaps, which the earlier stages retain better than the other types; we therefore also report recall by type and scores without research gaps.
- **Published methods.** Reproductions of published extraction methods on our splits are not yet included.
- **Cost.** Cost is measured as LLM tokens and call time on our hardware, not in money, and excludes Stages A and B.

6. Conclusions

We presented a three-stage funnel that extracts author-stated research gaps as unmodified source sentences. Rules remove text unlikely to contain gaps, a zero-shot NLI

classifier selects candidates without gap-labelled training data, and an LLM filter removes false positives in one of three modes set by a single instruction.

The stages have distinct roles. Stage A helps when sections are informative, Stage B sets the recall ceiling, and Stage C supplies most of the precision gain. The modes shift precision and recall in the intended direction on every dataset. In high-precision mode, precision is 66.2% at 84.9% recall on SAL-40 and 76.5% at 75.5% recall on FWS. On our gold corpus, whose labels come from LLMs and are not yet checked by people, 86.6% of kept sentences match those labels at 57.4% recall. Some gold test candidates influenced funnel design, so these results are provisional.

With the 27B LLM, the funnel has higher observed F_1 than whole-paper prompting on SAL-40, but the confidence interval includes zero and only the funnel receives section labels. It also has higher F_1 on FWS, but a broader baseline prompt favours the funnel. Whole-paper prompting has higher observed F_1 on arXiv papers against labels made from whole-paper inputs. With the 9B and 4B LLMs, the funnel is substantially better on arXiv, although the 9B intervals exclude zero only in balanced mode. The two same-generation smaller models show that the pattern is not unique to 4B, but the 27B checkpoint is from a newer model generation, so size remains confounded with model design.

For Stage B, the training-free NLI classifier comes within 0.021 F_1 of a supervised PubMedBERT in high-precision mode on SAL, but is 8.4 points less precise. Fine-tuning on LLM labels improves agreement with those labels, not with the human-labelled benchmarks.

The next steps are the human validation of the gold corpus, comparisons with published extraction methods, and the use of the extracted gaps for automated research-idea generation.

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Abbreviations

The following abbreviations are used in this manuscript:

CI	Confidence interval
FWS	Future work sentence (corpus)
LLM	Large language model
NLI	Natural language inference
NLP	Natural language processing
RCT	Randomized controlled trial
SAL	Self-acknowledged limitations (corpus)

Appendix A. Stage C prompt

The system prompt used on the gold corpus is reproduced below. The three modes differ only in the paragraph marked [MODE], whose three versions are listed after the prompt. For SAL and FWS, the decision rule keeps only item (a) or item (b), respectively; the other two items are added to the list of sentences to drop, and the words "gap" and "research gap" are replaced by "limitation" or "future-work direction". All prompts are in the code repository (`scripts/bench/stage_c_rubric_val.py`, `stage_c_sal40_modes.py`, `nli_sal_fws.py`).

> You are a precision filter for a research-gap extraction pipeline. Judge each candidate sentence: does it genuinely state a research gap?
>
> Each item has: "i" (integer id), "type" (the predicted gap type), "sentence" (the candidate to judge), and "context" (surrounding text). Judge the "sentence", using "context" only to disambiguate.
>
> Decision — answer YES (keep) if the sentence states ANY of: (a) a limitation / weakness / caveat / bias / constraint / assumption of the authors' OWN work (e.g., small sample, no blinding, single setting, short follow-up, confounding, limited generalizability, unvalidated measure, "we did not / could not …", "a limitation is …"); (b) an explicit FUTURE-WORK direction the authors propose ("in future work we will …", "future research should …", "it would be interesting to …"); (c) an unsolved problem / missing capability / open research gap in the field or prior work ("it remains unknown …", "no existing method …", "current approaches struggle with …").
>
> Answer NO (drop) if the sentence is instead: a result or finding, a method/experimental-setup description, a positive or contribution claim ("we propose/show/achieve …"), background or related-work description, a citation or reference fragment, acknowledgements / funding / author contributions / competing interests, a figure or table caption, a dataset/metric definition, or a generic non-gap statement.
>
> [MODE]
>
> Return JSON of the form {"results":[{"id":<i>, "keep":<bool>}, …]} with EXACTLY one entry per input item, where id is the item's "i" and keep=true means YES.

The [MODE] paragraph:

- **High recall:** "Be KEEP-BIASED: when you are genuinely uncertain whether a sentence is a real gap, answer YES. Only answer NO when it clearly is NOT a gap by the rules above."
- **Balanced:** "Judge each sentence on the rules above, without favouring YES or NO."
- **High precision (main mode):** "Be STRICT. Answer YES only when the sentence itself explicitly states the limitation, the proposed direction, or the unsolved problem. Also answer NO for: a general property or difficulty of a method or domain given as background or motivation; a statement of why the authors' approach is needed; an observed experimental outcome; and a passing remark that names no specific gap. When genuinely uncertain, answer NO."

Whole-paper and one-call-per-paper prompts. The whole-paper baseline and the one-call-per-paper funnel use the same prompt with some sentences replaced. The whole-paper prompt also replaces the opening line, which reads "You are reading sentences from scientific papers. Judge each sentence:". The description of the items is replaced by a description of the numbered input ("one per line as [id] sentence"; for the one-call-per-paper funnel, "the candidate sentences of ONE scientific paper … Sentences between candidates were removed"), and the output line asks for {"positives":[<id>, …]}, the ids of every sentence answered YES, in increasing order. The exact texts are in `scripts/bench/stage_c_per_paper_ft.py` and in the frozen plan of the whole-paper baseline (`docs/experiments/baseline_qwen/prompts/`).

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